

Unit 14, Lesson 5: Calculating Theoretical Probability of Compound Events

Benchmark

MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment.

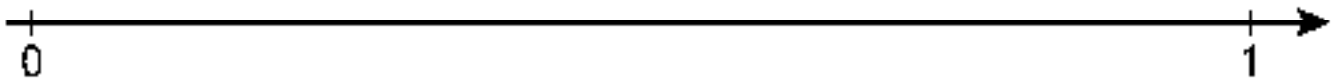
Learning Target

☐ I can calculate theoretical probability of compound events.

14.5.1 Warm-Up: Ordering Probabilities

1. Locate and label these numbers on the number line.

- A. 0.5
- B. 0.75
- C. 0.33
- D. 0.67
- E. 0.25

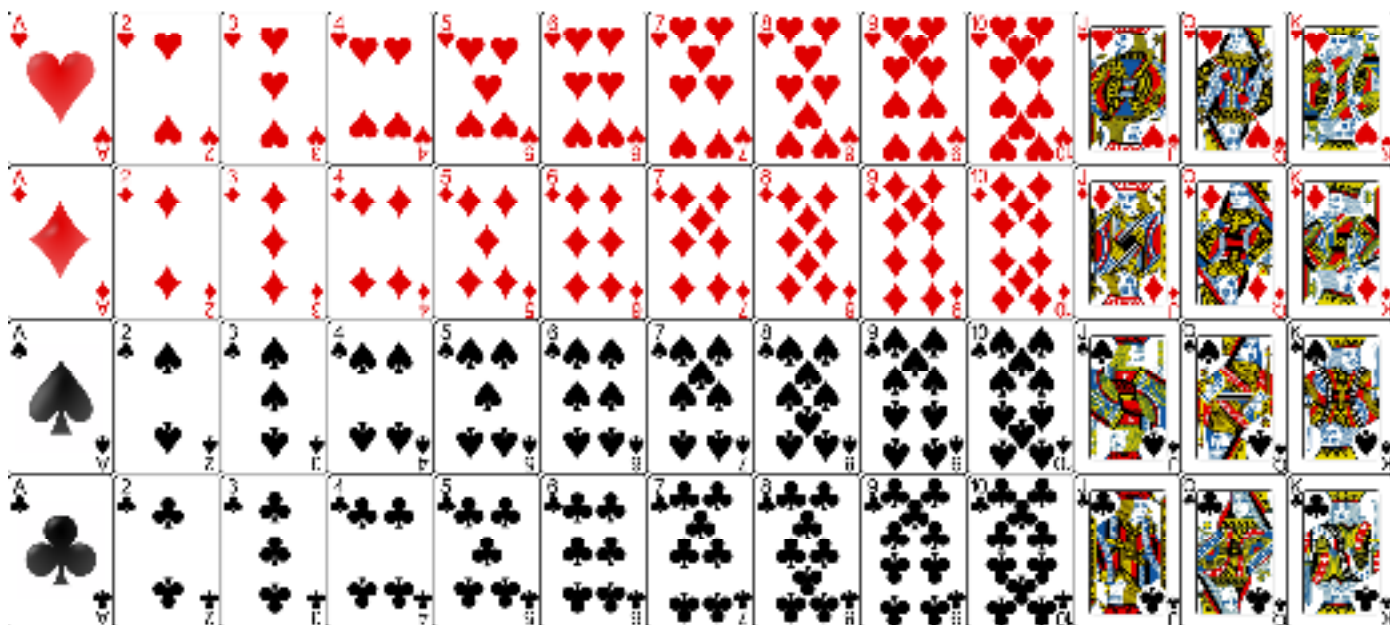


2. Choose one of the numbers from the previous question. Describe a game in which that number represents your probability of winning.

14.5.2 Activity: Make Your Own Game

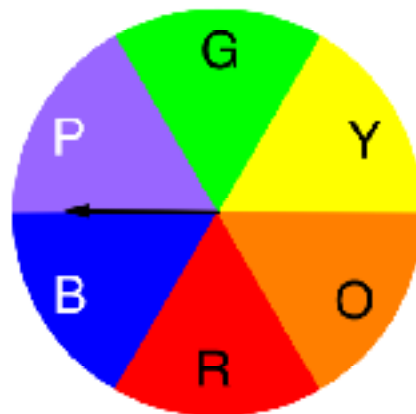
Option 1:

A card is drawn from a standard deck of 52 playing cards. There are 52 cards in a standard deck of cards; half are red (hearts and diamonds), half are black (spades and clubs), and 12 are "face" cards (jacks, queens, and kings). Aces are considered to have a value of one.



Option 2:

The fair spinner, shown, is spun.



Option 3:

A fair 12-sided die is rolled.



Work with your partner to complete the following.

Step 1: Choose any two simple experiments from the options provided.

Step 2: Create a game with your partner using the two options you chose.

- 1. Describe the game you created.**

- 2. Determine the number of outcomes in the sample space.**

- 3. Define Event A = the outcomes resulting in a win.**

Event A =

- 4. How many outcomes are in Event A?**

- 5. List the outcomes in Event A.**

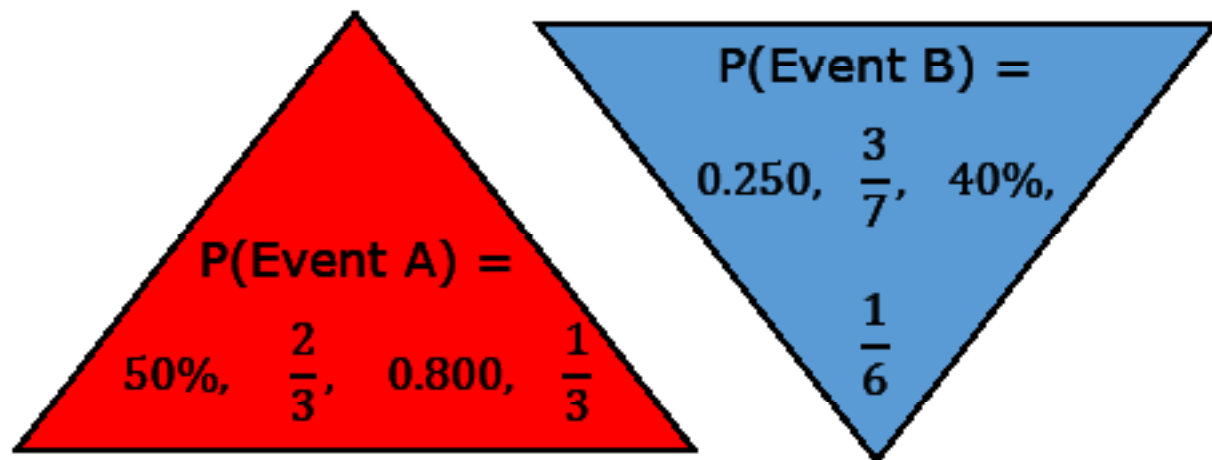
- 6. What is the probability of winning your game?**

- 7. Switch games with another partner pair. Verify their work.**

- 8. Make any adjustments necessary to your game based on the other partner pair's review of your game.**

14.5.3 Guided Practice: Calculating Probability

1. Choose a probability of Event A from the P(A) triangle. Choose a probability of Event B from the P(B) triangle. Use those probabilities to calculate P(A and B). In your work, use P notation to specify which probability you are using: P(A), P(B), and P(A and B).



2. Suppose you have an evenly divided spinner that shows all the days of the week. You also have a bag of papers that list the months of the year.

Suppose you spin the spinner and pick a paper from the bag.

What is the theoretical probability that you choose the paper with the current month and the spinner lands on the current day of the week? Use P notation when calculating the probability.

14.5.4 Try It: Calculating Probability

1. As part of a game at the school carnival, the player picks a marble from a bag, replaces the marble, and picks a marble from the bag again. The bag of marbles has two green, two red, and two blue marbles.
 - a. Calculate the theoretical probability that a player chooses a green marble and a blue marble. Write the probability as a fraction, as a decimal rounded to the nearest thousandth, and as a percent rounded to the nearest tenth of a percent.

$$P(\text{green and blue}) =$$

- b. Calculate the theoretical probability that a player doesn't pick a red on either of their picks.

$$P(\text{no red}) =$$

2. An integer chip with one yellow side and one red side is tossed 15 times.
 - a. What is the probability that the first toss results in red and all subsequent tosses result in yellow?
 - b. What is the probability that all 15 tosses result in the same color?

14.5.5 Your Turn!

1. Although she would like to have more, Wanda is only allowed to have two pets total. She is allowed to choose from these three options: a dog, a cat, and a rabbit. Wanda has decided to use a die to randomly choose which two pets she will have. If she rolls a 1 or a 2, she will get a dog, if the die lands on a 3 or 4 she will get a cat, and she will get a rabbit if the die lands on 5 or 6.

Calculate the theoretical probability that Wanda will end up with two cats.

2. A textbook has exactly 428 pages numbered in order starting with 1 and ending with 428. You flip to a random page in the book in a way that it is equally likely to stop at any of the pages. Then you repeat the process.

Consider Event N, turning to an odd numbered page three times in a row.

Calculate the theoretical probability of Event N.

14.5.6 Wrap-Up: Reflection

1. What was the **muddiest** (most difficult or confusing) point in today's lesson on calculating theoretical probability?



